ICCV HONOLULU HAWAII
OCT 19-23, 2025

Foundation Models in Visual Anomaly Detection: Advances, Challenges, and Applications

Organizers: Jiawen Zhu, Peng Wu, Olga Fink, Chengjie Wang, Guansong Pang

Guest Speakers: Ismail Nejjar, Jinlong Peng

ICCV 2025 Tutorial
Oct. 19, 2025, 9:00 AM - 12:00 PM (PDT)

SMU SINGAPORE MANAGEMENT UNIVERSITY

西北工業大學 NORTHWESTERN POLYTECHNICAL UNIVERSITY

EPFL

ICCV HONOLULU HAWAII
OCT 19-23, 2025

Tutorial on Datasets and Evaluation Protocols for Industrial Anomaly Detection

Chengjie Wang, Jinlong Peng, Bin-Bin Gao

Oct. 19, 2025

Agenda

- Datasets
  - 2D AD
  - Involving 3D
  - Involving Language
- Evaluation Protocols
  - Only Normal Data
  - Relying on Few Data
  - More Challenging Scenarios
  - Others
- Trends

Agenda

- Datasets
  - 2D AD
  - Involving 3D
  - Involving Language
- Evaluation Protocols
  - Only Normal Data
  - Relying on Few Data
  - More Challenging Scenarios
  - Others
- Trends

Datasets: 2D AD

- Overview

✓ MVTec AD, CVPR 2019

✓ VisA, ECCV 2022

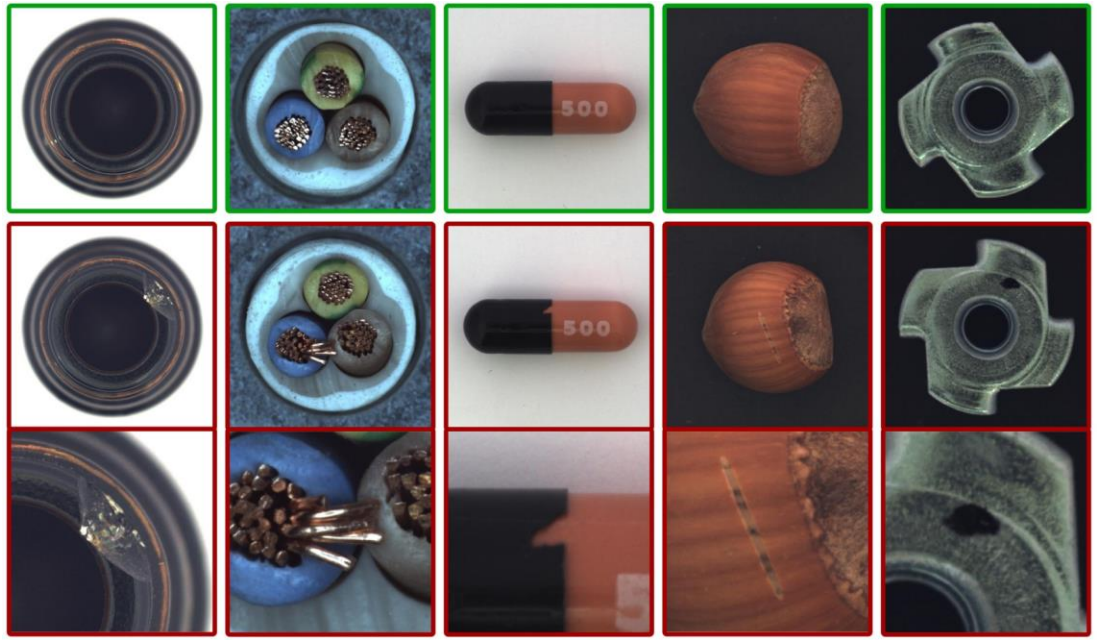
✓ Real-IAD, CVPR 2024

✓ MANTA, CVPR 2025

✓ Others: BTAD, SSGD, and 3CAD

Datasets: 2D AD

- MVTec AD, CVPR 2019

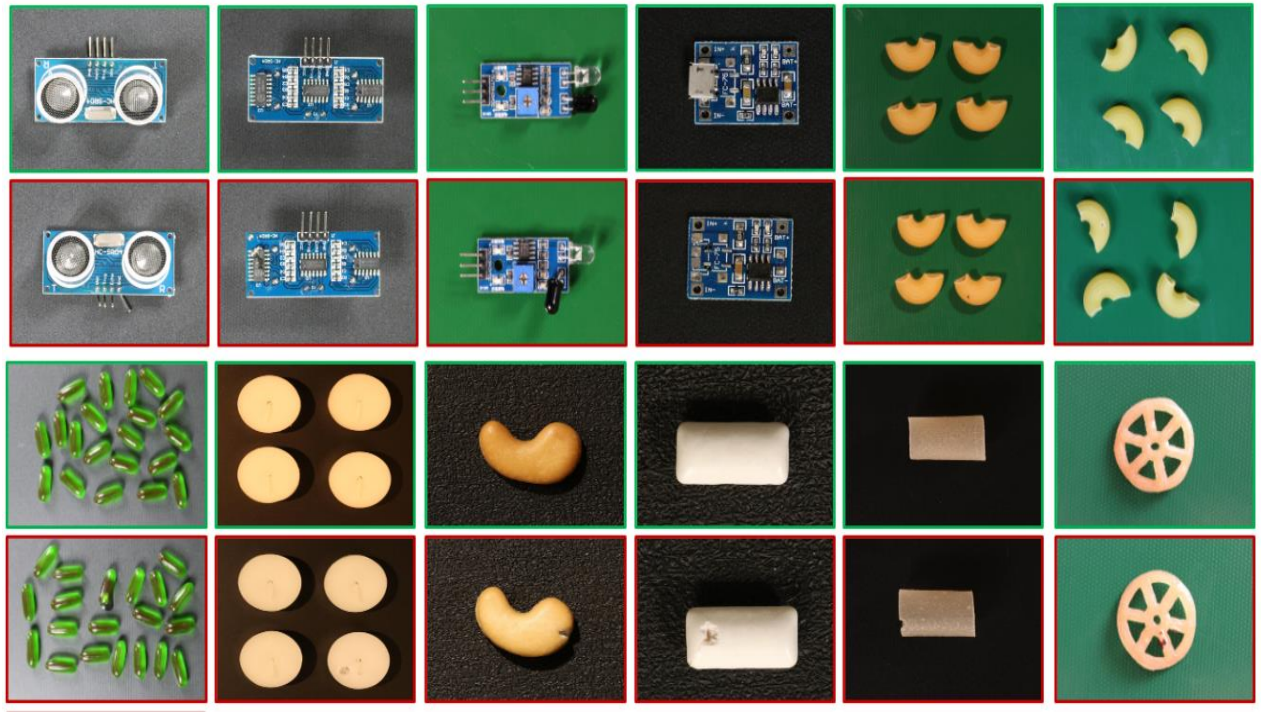


- a real-world dataset
- the most widely used AD dataset
- 15 categories
- 4096 normal images and 1258 anomaly images

- P. Bergmann et al. Mvtec ad–a comprehensive real-world dataset for unsupervised anomaly detection. CVPR 2019

Datasets: 2D AD

- VisA, ECCV 2022

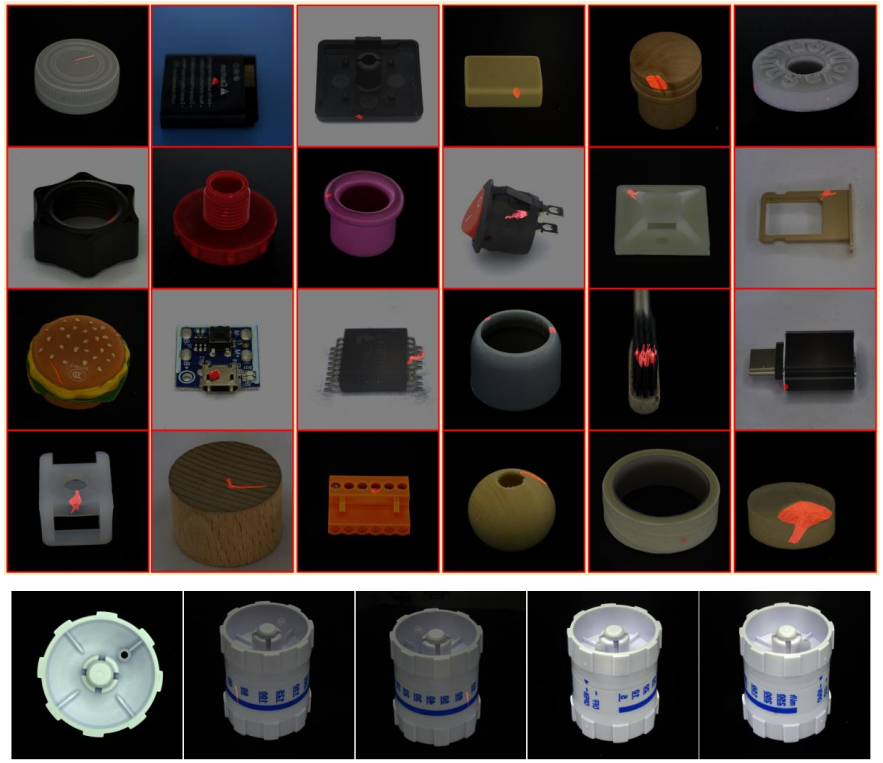


- A real-world dataset
- multi-scenarios: complex structure, multiple instances, single instance
- 12 objects
- 9,621 normal images and 1,200 anomaly images

- Y. Zhou et al. Spot-the-difference self-supervised pretraining for anomaly detection and segmentation. ECCV 2022

Datasets: 2D AD

- Real-IAD, CVPR 2024



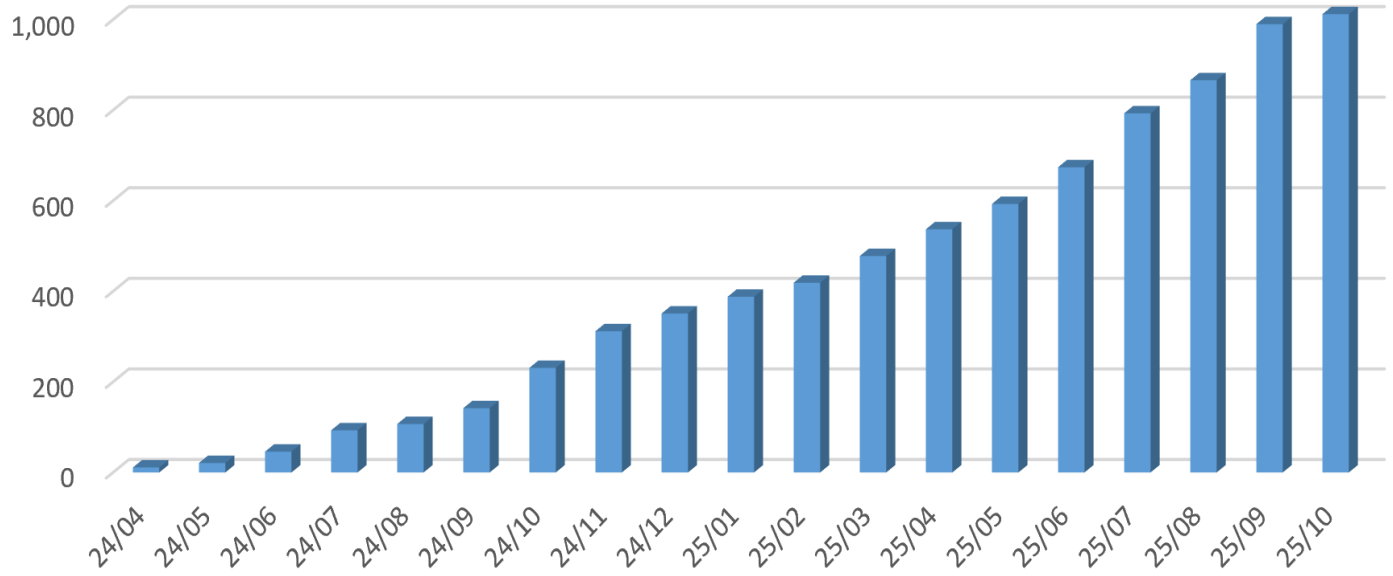
- a real-world dataset
- the first large scale AD dataset, 10x larger than previous datasets
- the first multi-view AD dataset
- 10 categories
- 99,721 normal images and 51,329 images
- high image resolution: 2000~5000

- **C. Wang** et al. Real-IAD: A real-world multi-view dataset for benchmarking versatile industrial anomaly detection. CVPR, 2024

Datasets: 2D AD

- Real-IAD (cont.)

The growth trend of **Hugging Face user** counts for the Real-IAD dataset

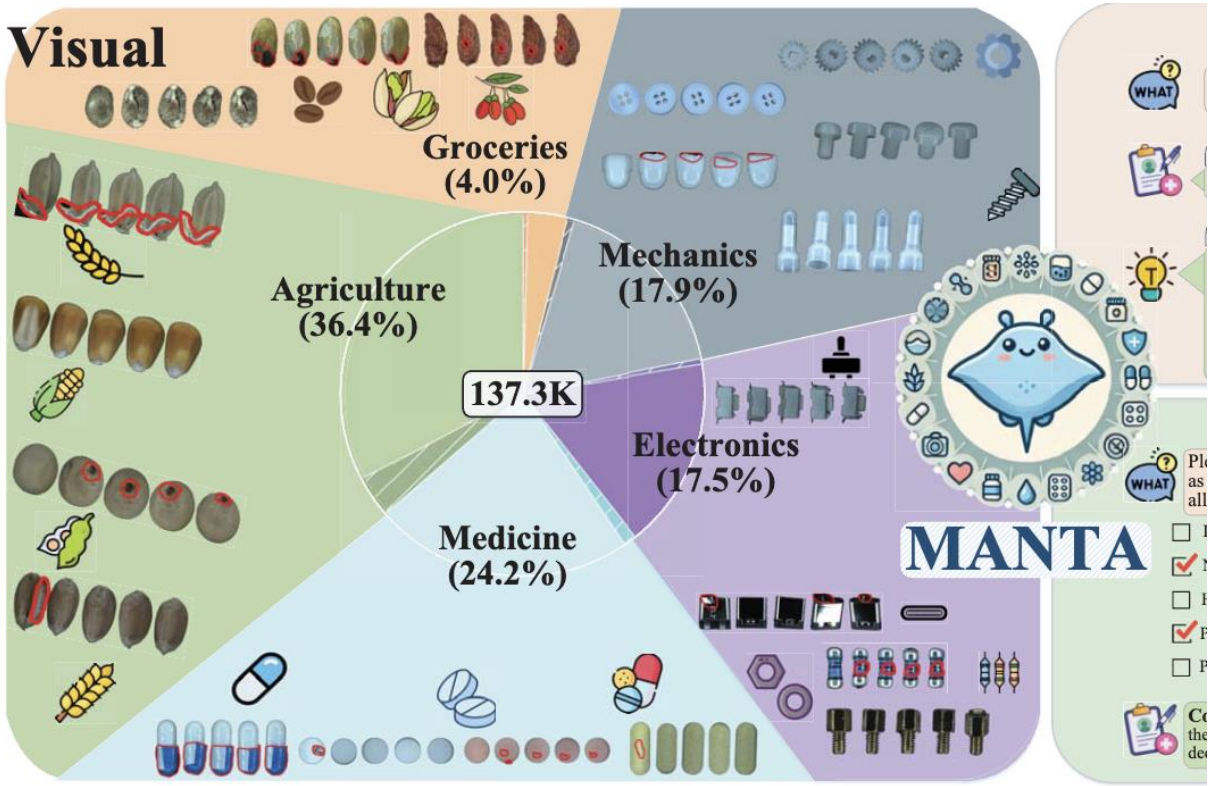


- quick spread
- 35 countries and regions
- exceeding 1,000+ users
- about 200+ universities
- about 180+ enterprises

- **C. Wang** et al. Real-IAD: A real-world multi-view dataset for benchmarking versatile industrial anomaly detection. CVPR, 2024

Datasets: 2D AD

- MANTA



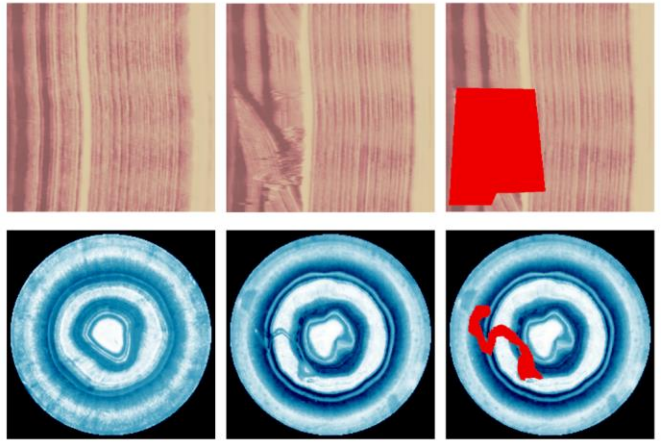
- a real-world dataset
- multi-view: five distinct viewpoints
- multi-domain: agriculture, medicine, electronics, mechanics, and groceries
- anomalies in tiny objects
- 38 categories
- 686,690 images

- L. Fan et al. MANTA: A Large-Scale Multi-View and Visual-Text Anomaly Detection Dataset for Tiny Objects. CVPR, 2025

Datasets: 2D AD

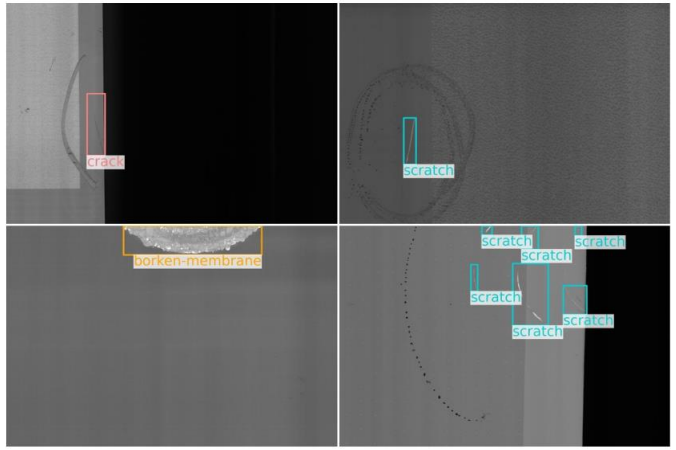
- Others

BTAD, ISIE 2021



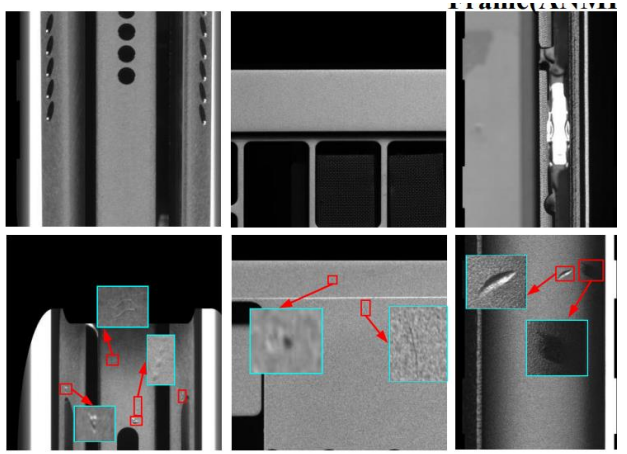
3 categories, 1344 total images

SSGD, ICASSP 2023



A smartphone screen glass dataset: 2,504 anomaly images

3CAD, AAAI 2025



3C products: phone, pad, laptop
About 15,000 normal images and 11,000 anomaly images

- P. Mishra et al. Vt-ADL: A vision transformer network for image anomaly detection and localization. ISIE 2021
- H. Han et al. SSGD: A smartphone screen glass dataset for defect detection. ICASSP 2023
- E. Yang et al. 3CAD: A Large-Scale Real-World 3C Product Dataset for Unsupervised Anomaly Detection. AAAI 2025

Datasets: AD involving 3D

- Overview

✓ MVTec-3D-AD, VISAPP 2022

✓ Real-3D AD, NeurIPS 2023

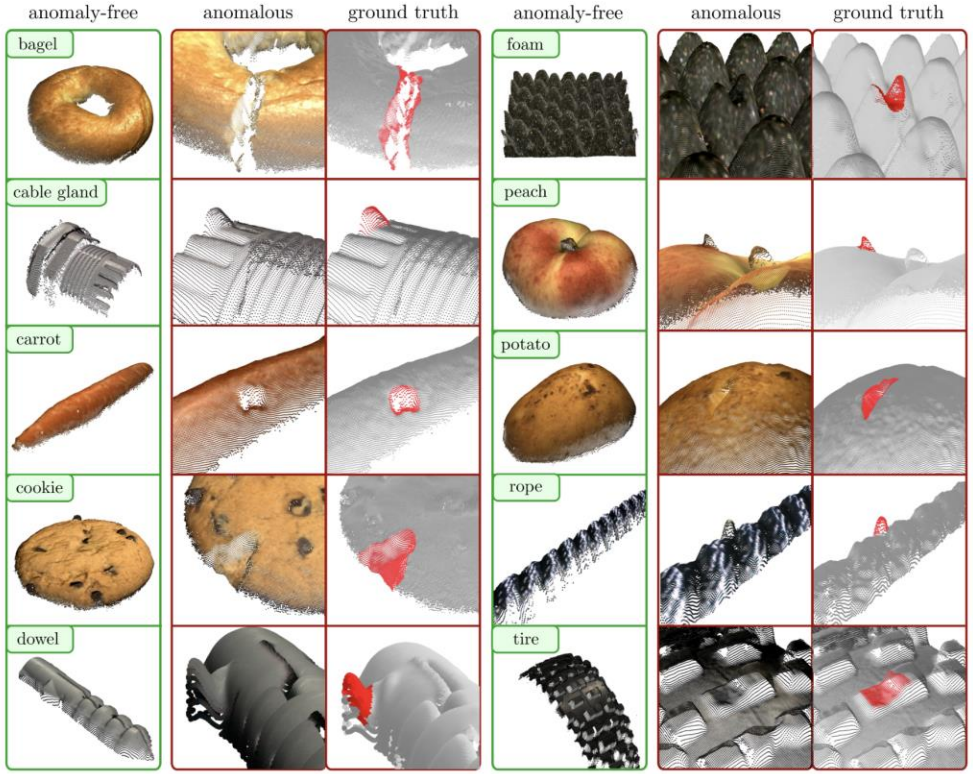
✓ Real-IAD D^3 , CVPR 2025

✓ MulSen-AD, CVPR 2025

✓ Others: Eyecandies and ShapeNet

Datasets: AD involving 3D

- MVTec 3D-AD, VISAPP 2022

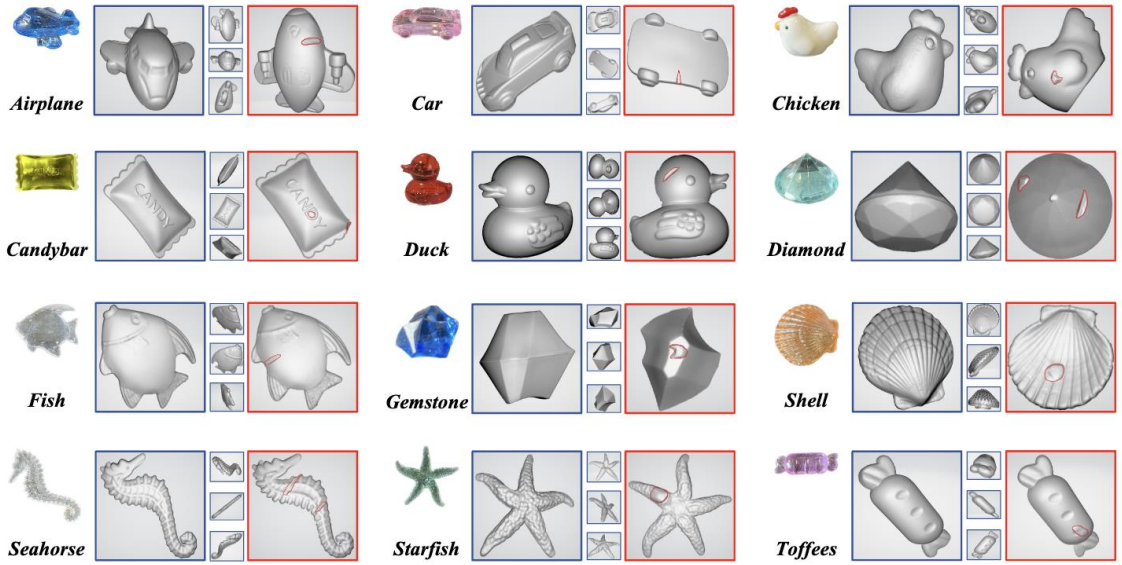


- a real-world 3D dataset
- the first comprehensive 3D AD dataset
- 30 categories
- 4147 high resolution images
- 3D point cloud + RGB

- P. Bergmann et al. The MVTec 3D-AD Dataset for Unsupervised 3D Anomaly Detection and Localization. VISAPP 2022.

Datasets: AD involving 3D

- Real-3D AD, NeurIPS 2023



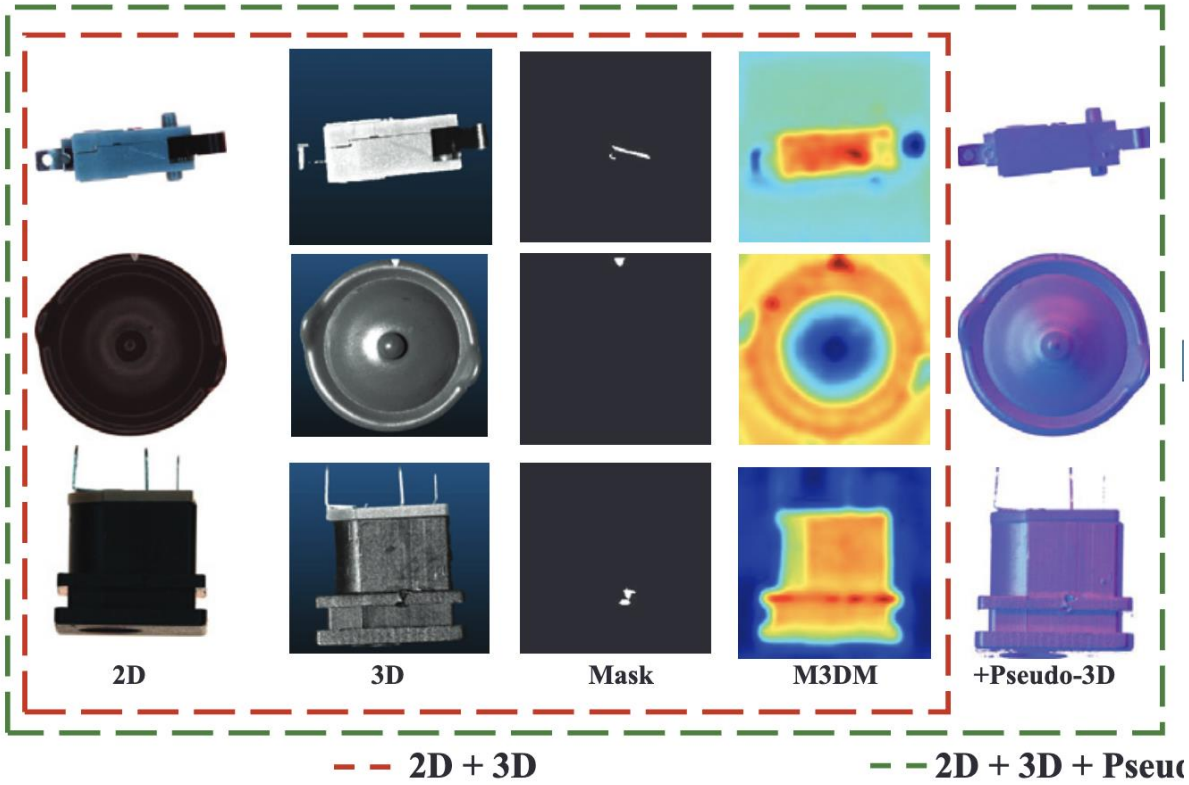
Scanner	PMAX-S130	Zivid One-Plus[a]
Dataset	Real3D-AD (Ours)	MVTec 3D-AD [1]
FOV	100cm to 400cm	60cm to 200cm
Point Precision	0.011mm-0.015mm	0.11mm
Spatial Distance	0.04mm-0.07mm	0.37mm
3D Format	ASC, PLY, STL, OBJ, IGES	TIFF

- a real-world 3D dataset
- high precision, no blind spot, and realistic
- 12 categories
- 652 normal images and 602 anomaly images
- 3D point cloud + RGB

- J. Liu et al. Real3D-AD: A Dataset of Point Cloud Anomaly Detection. NeurIPS 2023.

Datasets: AD involving 3D

- Real-IAD D^3 , CVPR 2025

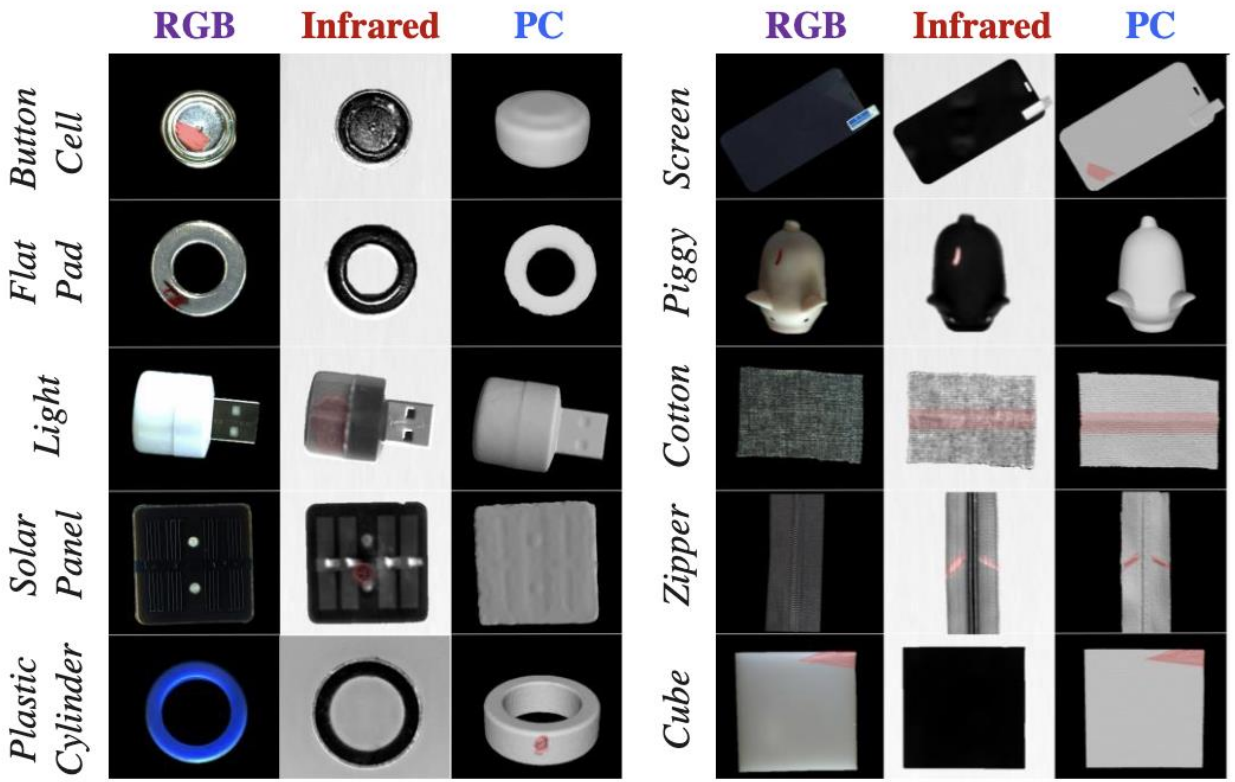


- a real-world pseudo-3D dataset
- finer defects, diverse anomalies, and greater scale
- pseudo-3D images generated through photometric stereo
- 20 categories
- 8,450 high-precision images
- 3D point cloud + RGB + pseudo-3D

- W. Zhu et al. Real-IAD D^3: A Real-World 2D/Pseudo-3D/3D Dataset for Industrial Anomaly Detection. CVPR 2025.

Datasets: AD involving 3D

- MulSen-AD, CVPR 2025



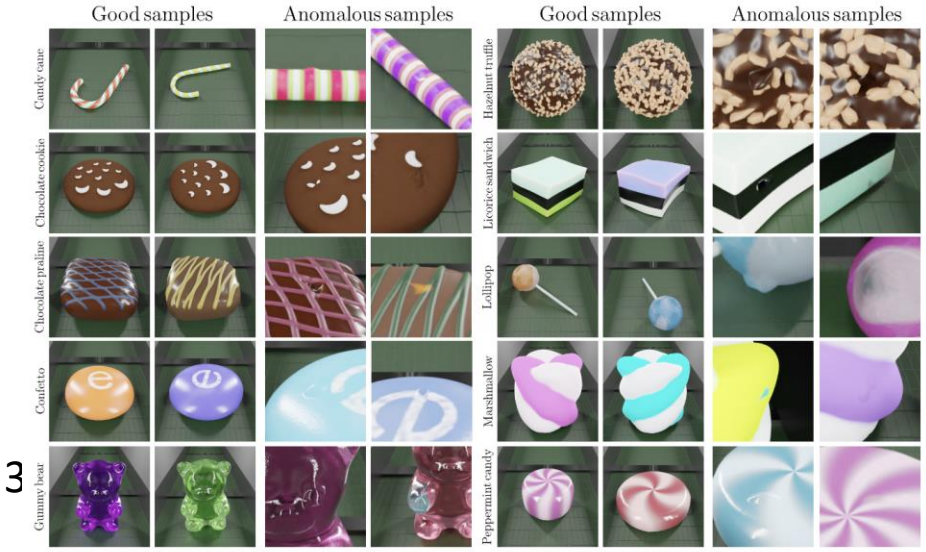
- the first real-world dataset for multi-sensor anomaly detection
- 15 categories
- 2035 samples
- RGB + IR + Point Cloud

- W. Li et al. Multi-Sensor Object Anomaly Detection: Unifying Appearance, Geometry, and Internal Properties. CVPR 2025.

Datasets: AD involving 3D

- Others： synthetic 3D datasets

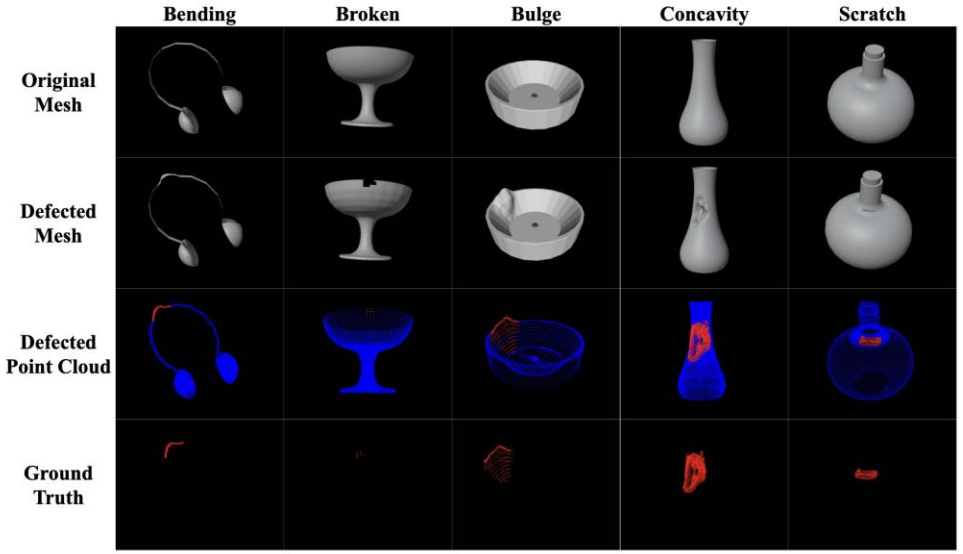
Eyecandies, ACCV 2022



10 categories， 15,000 samples

RGB + Depth + Normals 3D

ShapeNet, CVPR 2024



40 categories， 1600 samples

3D point clouds + Meshs

- L. Bonfiglioli et al. The Eyecandies Dataset for Unsupervised Multimodal Anomaly Detection and Localization. ACCV 2022.
- W. Li et al. Towards Scalable 3D Anomaly Detection and Localization: A Benchmark via 3D Anomaly Synthesis and A Self-Supervised Learning Network. CVPR 2024.

Datasets: AD involving Language

- Overview

✓ MVTec LOCO, IJCV 2022

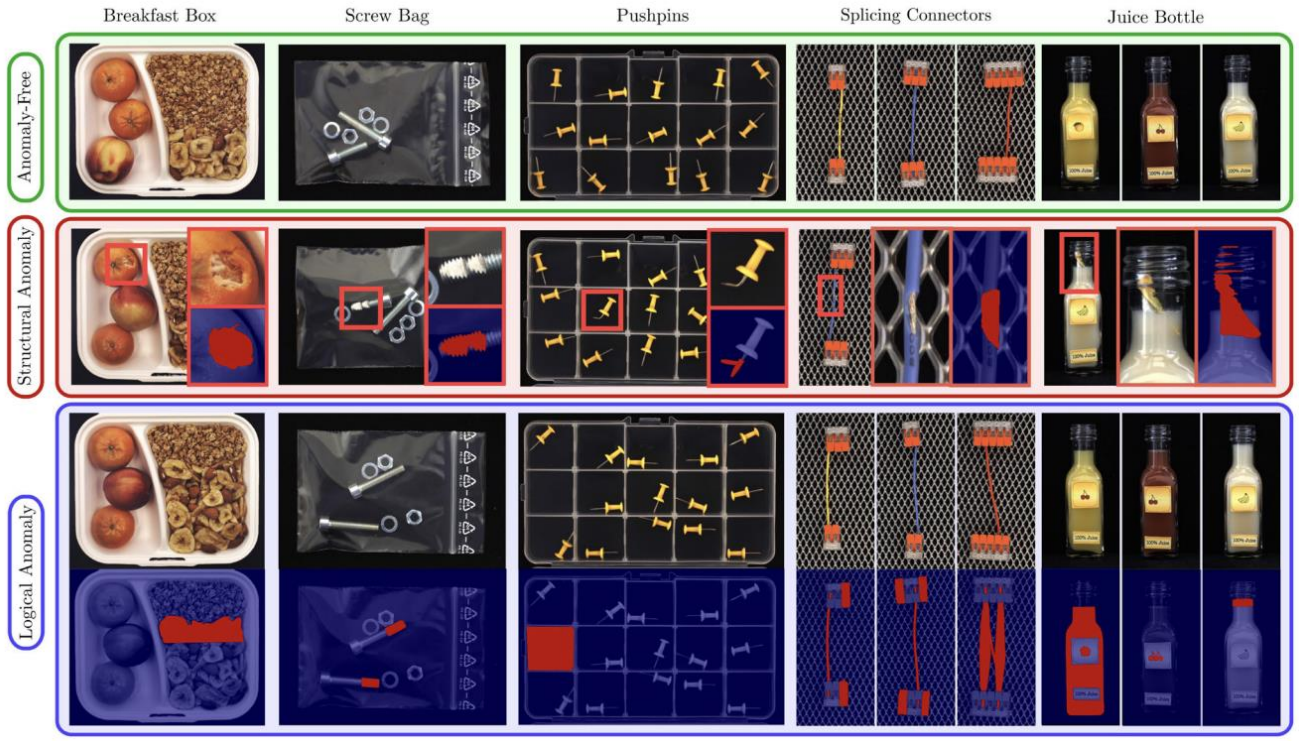
✓ MMAD, ICLR 2025

✓ Anomaly-Instruct-125k, CVPR 2025

✓ TriAD, ICCV 2025

Datasets: AD involving Language

- MVTec LOCO, IJCV 2022

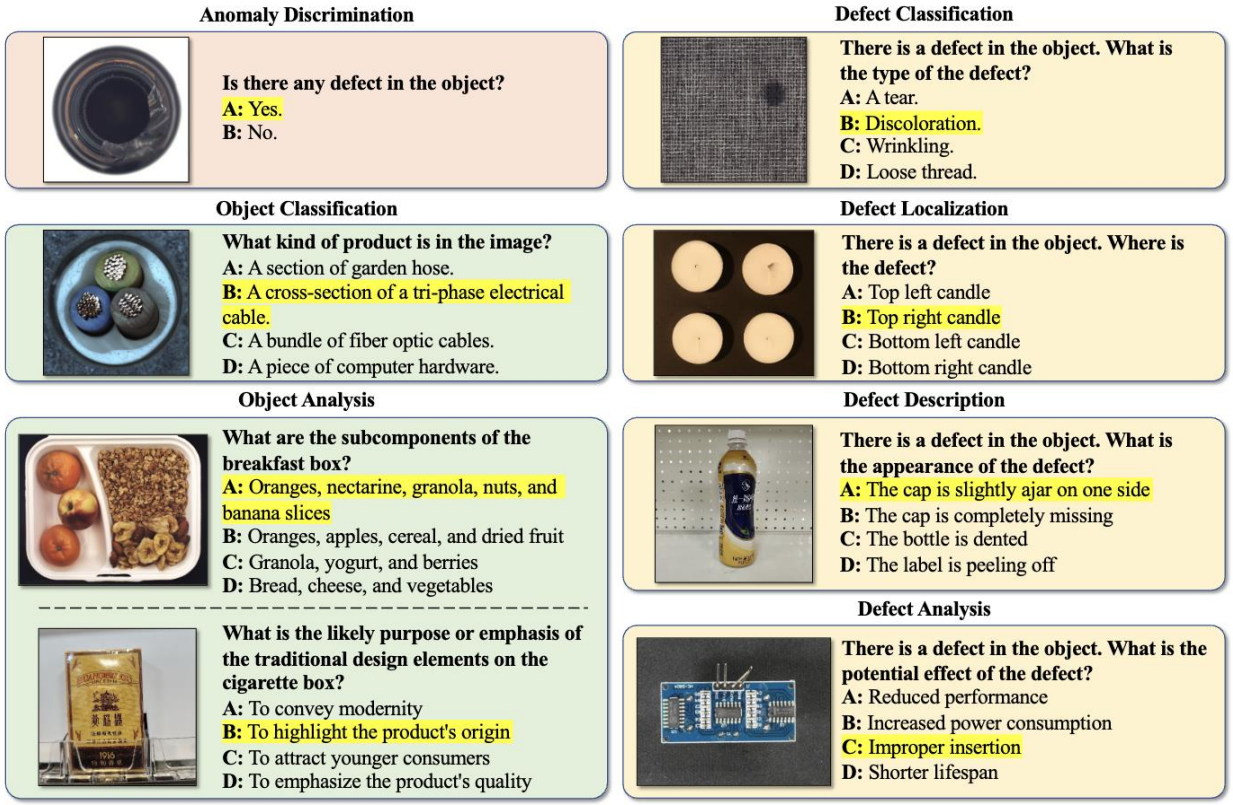


- a real-world dataset
- the first dataset covers both structural and logical anomalies
- saturated per-region overlap (sPRO) metric
- 5 categories
- 3,644 images

- P. Bergmann et al. Beyond Dents and Scratches: Logical Constraints in Unsupervised Anomaly Detection and Localization. IJCV 2022.

Datasets: AD involving Language

- MMAD, ICLR 2025

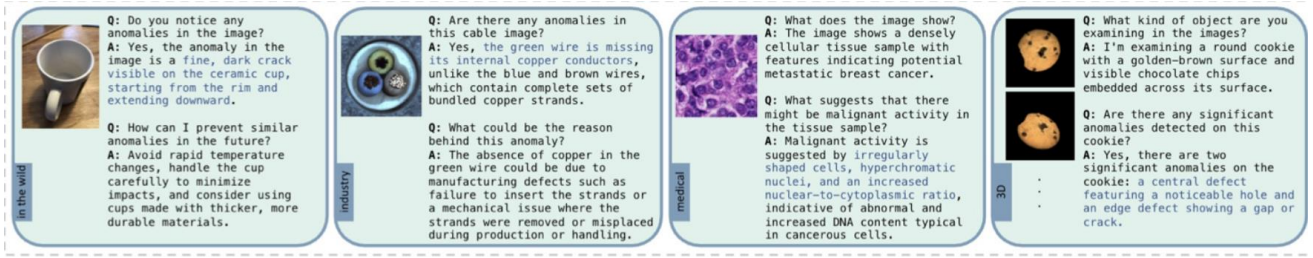


- the first evaluation benchmark for MLLMs in the task of IAD
- a novel pipeline to generate the MMAD dataset with GPT-4V
- 38 categories
- 8,366 images and 39,672 multi-choice questions

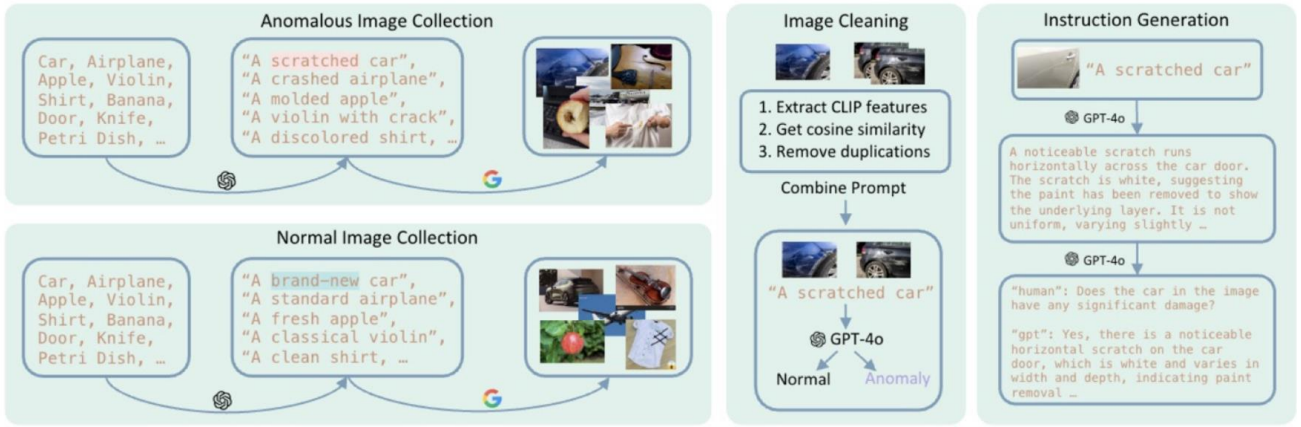
- X. Jiang et al. MMAD: A Comprehensive Benchmark for Multimodal Large Language Models in Industrial Anomaly Detection. ICLR 2025.

Datasets: AD involving Language

- Anomaly-Instruct-125k, CVPR 2025



We introduce **Anomaly-Instruct-125k**, which consists of four types of data (industrial, medical, 3D, and in-the-wild), for visual instruction tuning and **VisA-D&R** for benchmarking the performance on anomaly detection and reasoning.

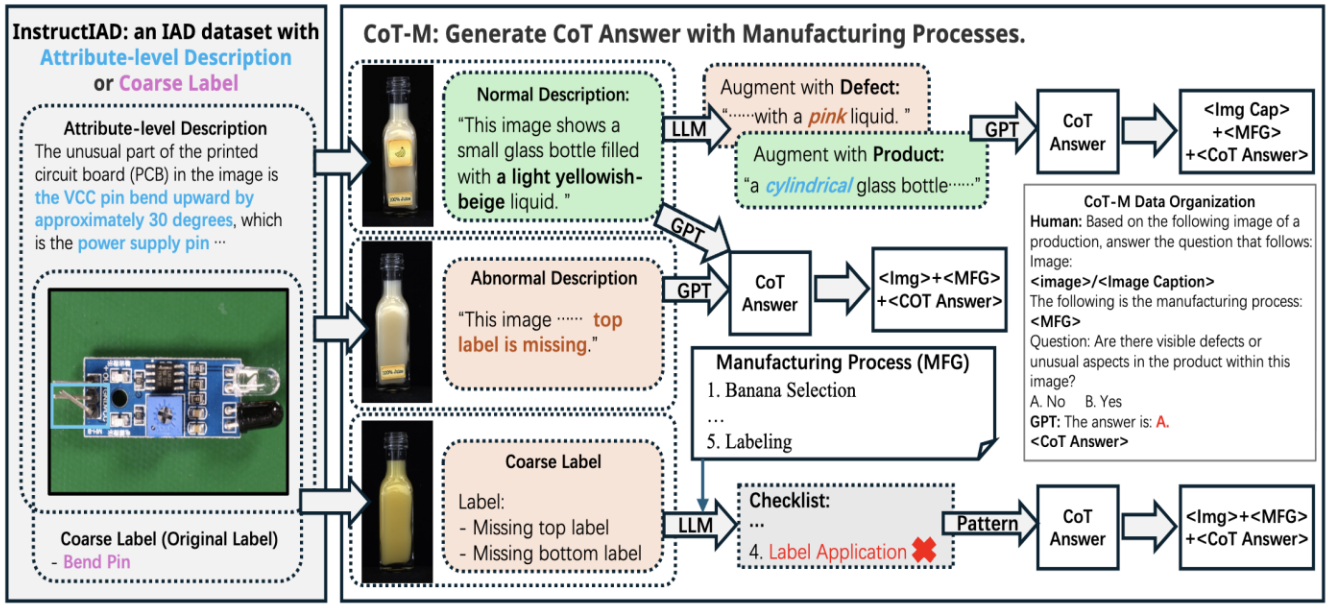


- the first visual instruction tuning dataset and benchmark for anomaly detection and reasoning
- cover both 2D to 3D data across industry to medical domains
- collect 72k wild images
- instructions format: multi-round conversations

- J. Xu et al. Towards Zero-Shot Anomaly Detection and Reasoning with Multimodal Large Language Models, CVPR 2025.

Datasets: AD involving Language

- TriAD, ICCV 2025



- an instruction-tuning dataset
- fine-grained, attribute-level captions
- support multi-task: anomaly detection, caption and analysis
- 40 categories
- 9,444 abnormal and 13,578 normal samples from Real-IAD and VisA

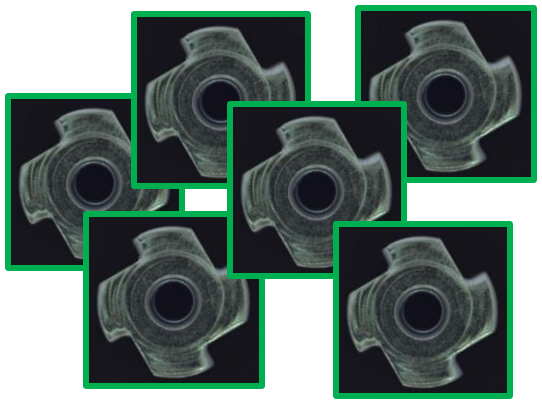
- Y. Li et al. Triad: Empowering LMM-based Anomaly Detection with Expert-guided Region-of-Interest Tokenizer and Manufacturing Process, ICCV 2025.

Agenda

- Datasets
  - 2D AD
  - Involving 3D
  - Involving Language
- Evaluation Protocols
  - Only Normal Data
  - Relying on Few Data
  - More Challenging Scenarios
  - Others
- Trends

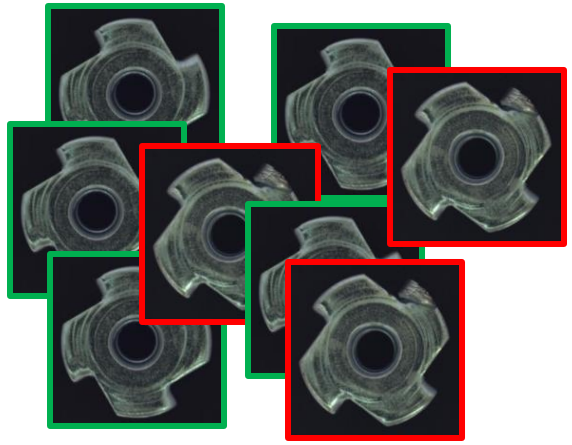
Evaluations: Only Normal Data

- UIAD (Unsupervised Industry Anomaly Detection)



Train set

Normal samples



Test set

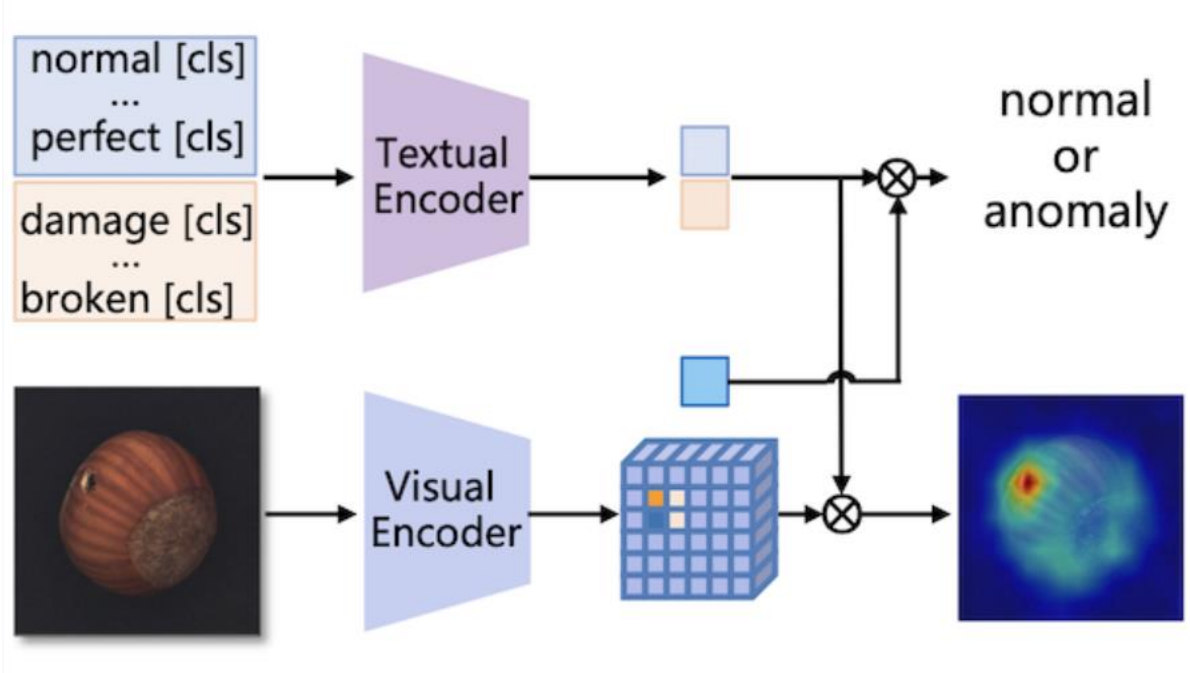
Normal & Abnormal

- one model for one class
- train set: only normal images
- test set: normal and/or anomaly images
- Representative methods
  - ✓ PatchCore, CVPR 2022
  - ✓ RD, CVPR 2022
  - ✓ SimpleNet, CVPR 2023

Evaluations: Relying on Few Data

- Zero-Shot AD

text prompts based CLIP model

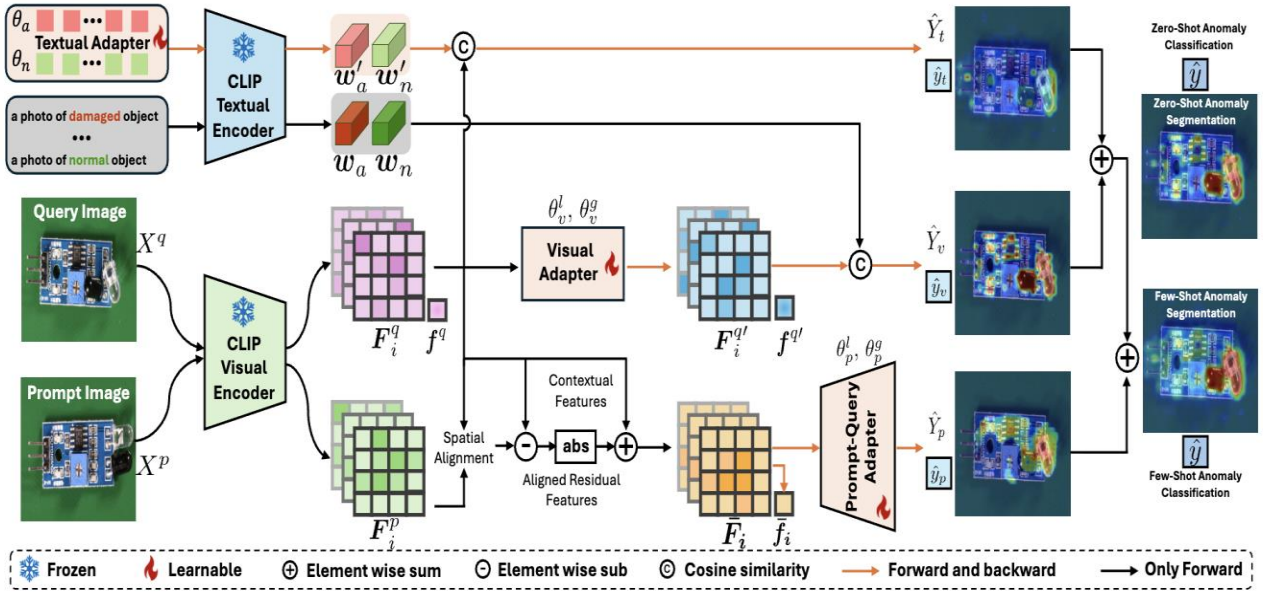


- one model for any class
- training-free on target domains
- representative methods
  - ✓ WinCLIP, CVPR 2023
  - ✓ AnomalyCLIP, ICLR 2024
  - ✓ AdaCLIP, VCPCLIP, ECCV 2024
  - ✓ AACLIP, ICCV 2025
  - ✓ AdaptCLIP, arXiv 2025

Evaluations: Relying on Few Data

- Few-Shot AD

text prompts and few-shot normal image prompts on CLIP model



- one model for any class
- only need few-shot normal images from target domains
- representative methods
  - ✓ WinCLIP+, CVPR 2023
  - ✓ InCtrl, CVPR 2024
  - ✓ MetaUAS, NeurIPS 2024
  - ✓ UniVAD, CVPR 2025
  - ✓ AdaptCLIP, arXiv 2025

Evaluations: More Challenging Scenarios

- Overview

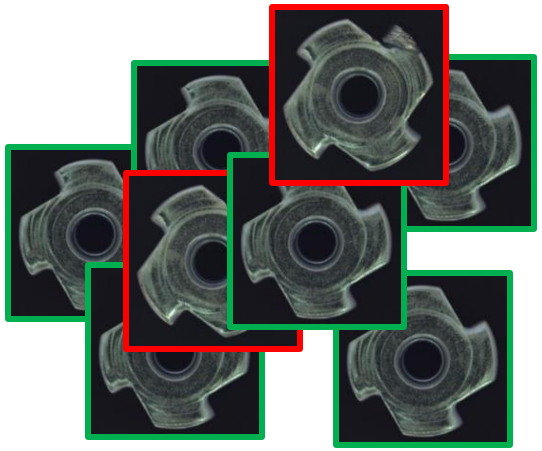
✓ FUIAD: from 2D to 3D

✓ Multi-Class AD

✓ Multi-View AD

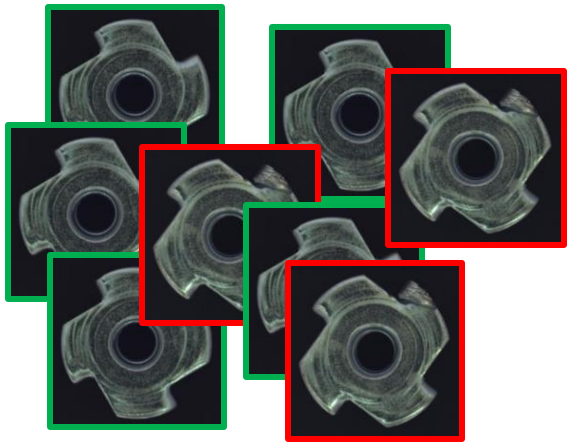
Evaluations: More Challenging Scenarios

- FUIAD (Fully UIAD / noisy AD)



Train set

Abnormal samples < 40%



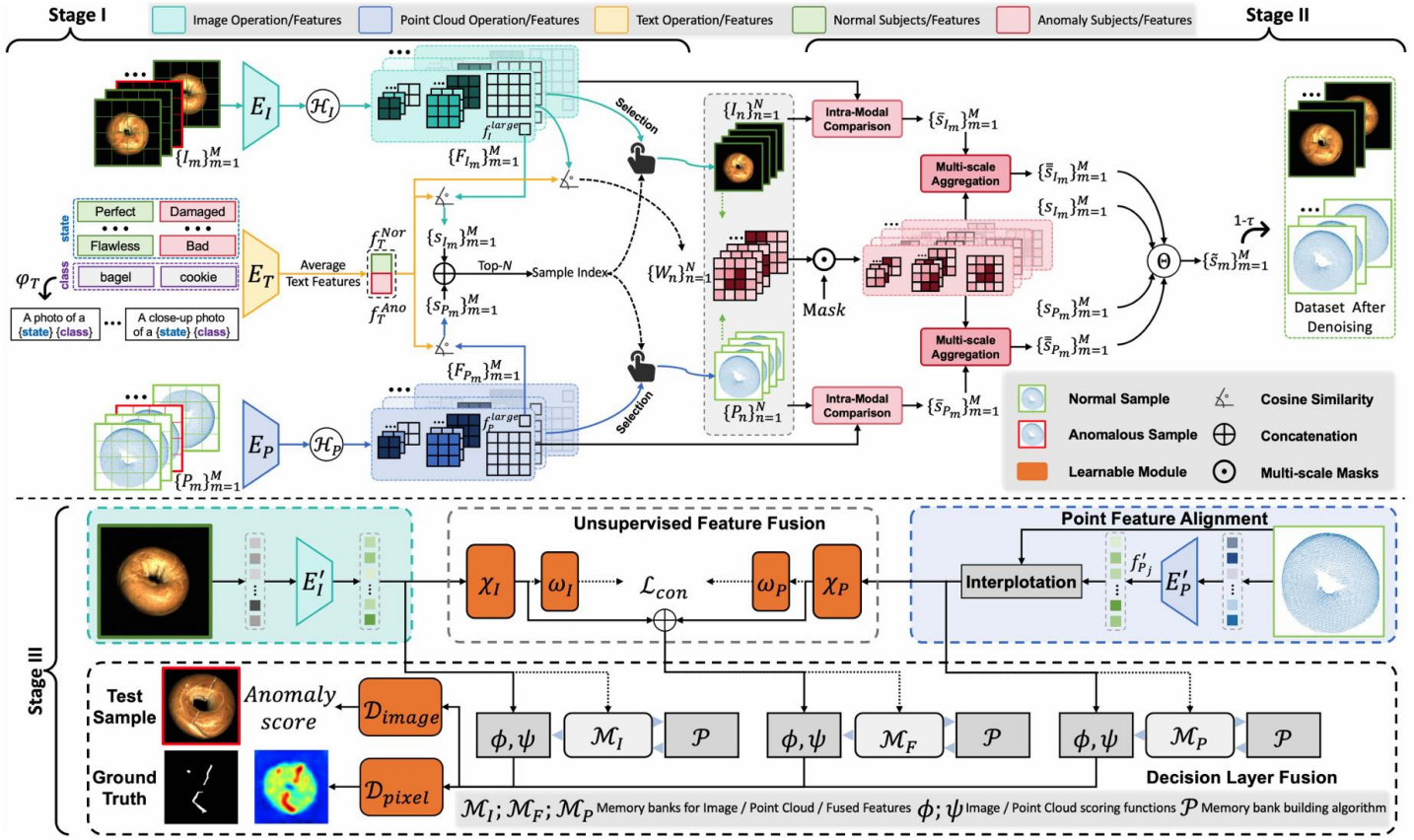
Test set

Normal & Abnormal

- directly train on production line data, without data cleaning or manual labeling
- denoising: NN, Gausssion or LOF
- representative methods
  - ✓ SoftPatch, NeurIPS 2022
  - ✓ SoftPatch+, PR 2024
  - ✓ FUN-AD, WACV 2024

Evaluations: More Challenging Scenarios

- FUIAD (2D/3D)



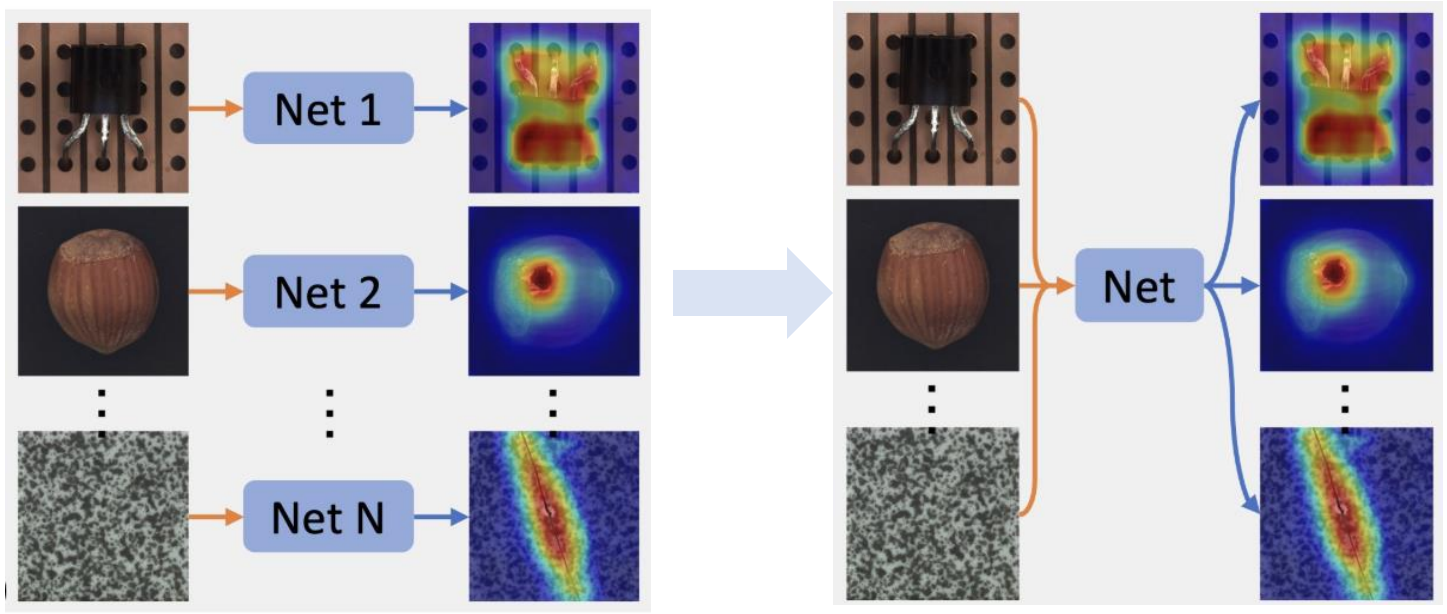
- representative methods

✓ M3DM-NR, TPAMI 2025

denoising: use anomaly maps of the reference samples as reference, and achieve denoising with image, point cloud, and text information

Evaluations: More Challenging Scenarios

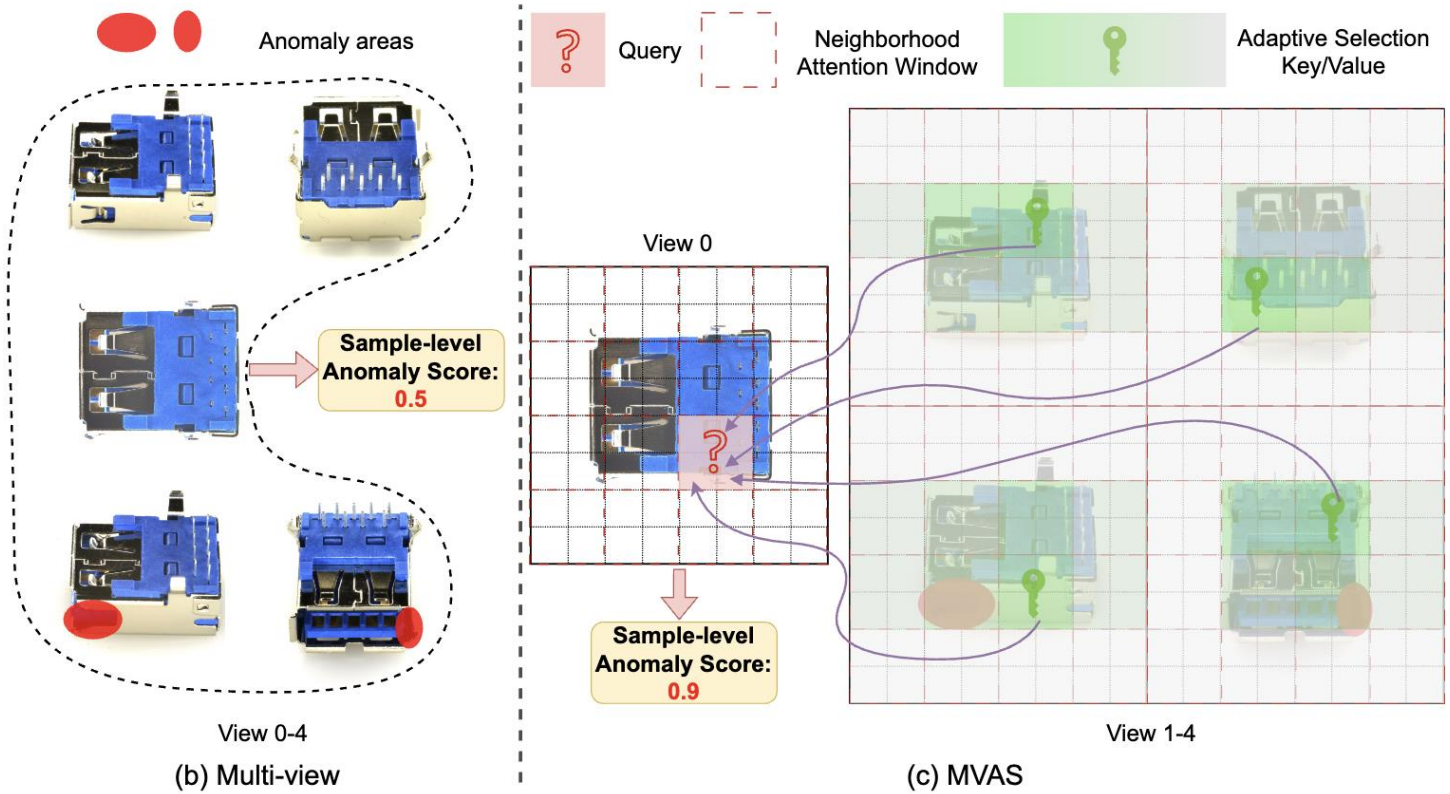
- Multi-Class AD



- one model for multi-class
- only normal images for training
- representative methods
  - ✓ UniAD, NeurIPS 2022
  - ✓ OneNIP, ECCV 2024
  - ✓ RLR, ECCV 2024
  - ✓ MambaAD, NeurIPS 2024
  - ✓ Dinomaly, CVPR 2025

Evaluations: More Challenging Scenarios

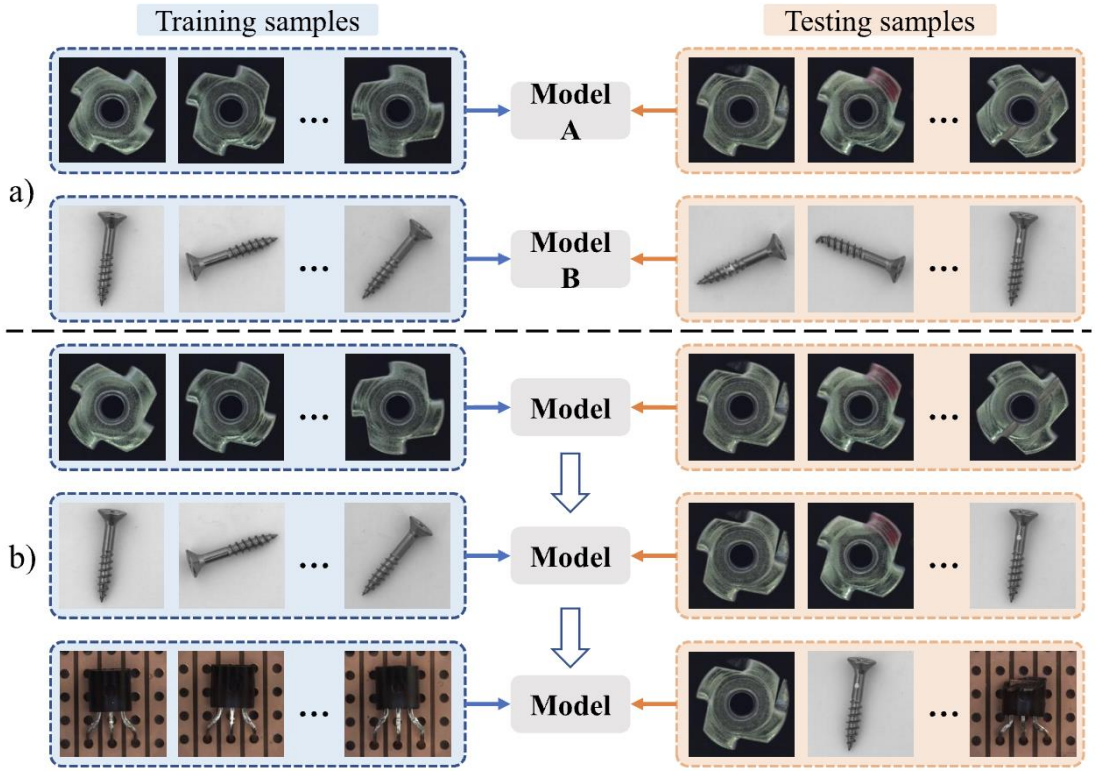
- Multi-View AD



- learns and integrates features from multi-views
- representative methods
  - ✓ Diffusion MVAD, ECCV 2024
  - ✓ MVAS, TMM 2025
  - ✓ IDIF, AAAI 2025

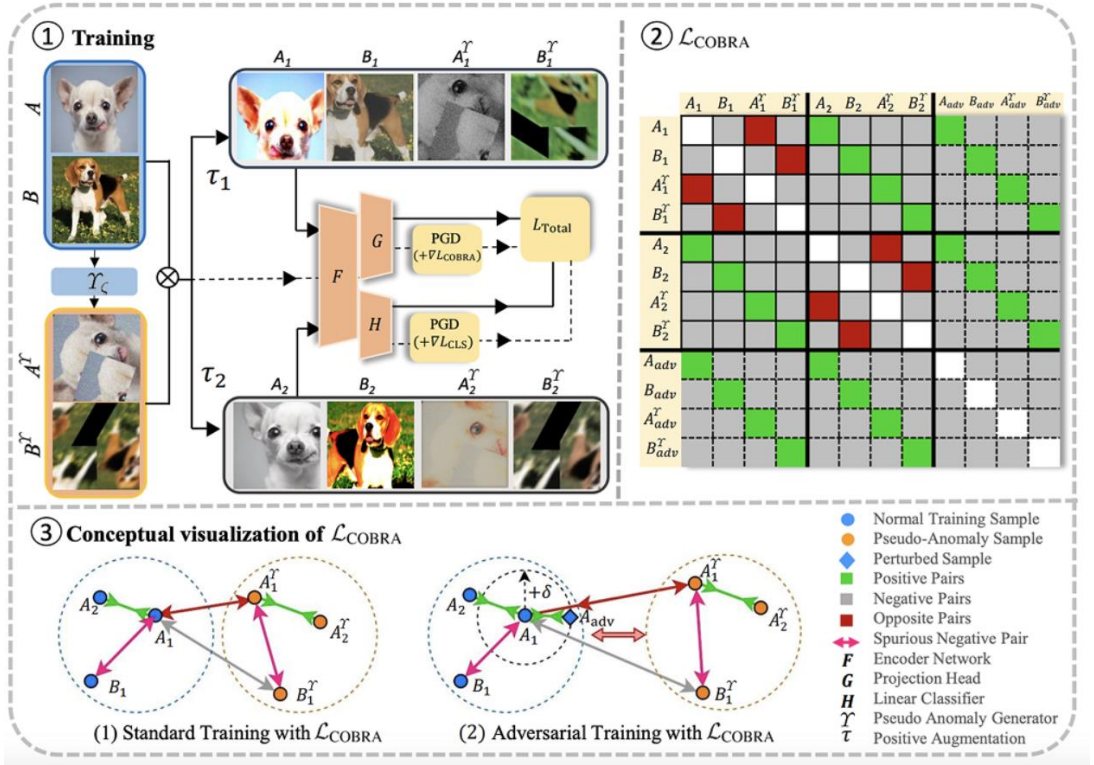
Evaluations: Others

- Continual AD



UCAD, AAAI 2025

- Adversary Attack



COBRA, ICLR 2025

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  - Involving Language
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Trends: IAD

- A Real-IAD view

✓ from UIAD to FUIAD

✓ from multi-class to large-scale multi-class

e.g., MVTec (15 classes), VisA (12 classes) —> Real-IAD (30 classes), Real-IAD Variety (160 classes)

✓ from single-view to multi-view

e.g., single-view MVTec and VisA —> multi-view Real-IAD, MANTA

✓ from single sensor to multi-sensor

e.g., single RGB —> RGB imaging, high-precision laser scanning, and lock-in infrared thermograph

✓ from unimodality to multimodality

e.g., pure vision —> vision-language

Trends: More than IAD

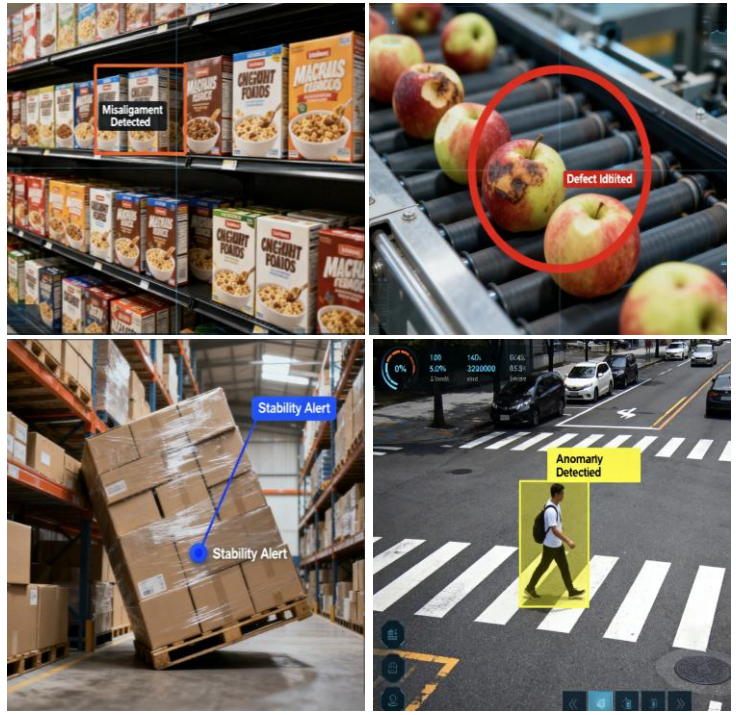
✓ from industrial to medical, even universal scenarios



industrial



medical



universal scenarios

Thanks for Your Attention

Q & A